MOVING RIGID LESLIE MODELS ON CONSECUTIVE LOGARITHMIC WINDOWS: CRITICAL SPECTRAL GEOMETRY, OVERLAP RIGIDITY, SCALAR INNOVATIONS, AND A CAUSAL FORECASTING TEST

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Remark 0.1. This document is a research notebook. The algebraic identities and local differential constraints are proved under explicit regularity and simplicity hypotheses. The proposed connection with consecutive windows of the Mertens function is an experimental programme, not a proved arithmetic theorem. No statement below proves a new estimate for the Mertens function or the Riemann hypothesis.

1. Purpose and change of viewpoint

The earlier static experiments asked whether one finite collection of fitted frequencies could be embedded in one nonnegative Leslie matrix. Allowing auxiliary eigenvalues made that static problem feasible in all forty reported configurations, but the result did not distinguish a coherent dynamical law from independent fitting.

The present formulation replaces independent static realizations by a sequence of rigid Leslie matrices attached to consecutive logarithmic windows. The object of study is therefore a non-autonomous product, or linear cocycle, rather than one fixed matrix. Linear cocycles and their long products are standard objects in dynamical-systems theory [2].

Warning 1.1. The use of a time-dependent matrix product is not itself a new result. The potentially discriminating content is the simultaneous enforcement of rigidity, nonnegativity, continuity through the feasible set, and prediction of a future window without recalibration.

2. Rigid moving Leslie matrices

Definition 2.1. For $m \geq 2$, a companion Leslie matrix is

$$L(a) = \begin{pmatrix} a_1 & a_2 & \cdots & a_m \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & & \ddots & 0 \end{pmatrix}, \qquad a_\ell \geq 0. \tag{2.1}$$

Its characteristic polynomial is

$$P(z;a) = z^m - \sum_{\ell=1}^{m} a_\ell z^{m-\ell}. \tag{2.2}$$

Fix $K \geq 1$ and set

$$m = 2K + 1. \tag{2.3}$$

For $b > 1$ and phases $\Omega = (\omega_1, \ldots, \omega_K)$, prescribe the roots

$$b, \qquad \sqrt{b}\, e^{\pm i\omega_1}, \ldots, \sqrt{b}\, e^{\pm i\omega_K}. \tag{2.4}$$

The unique monic polynomial with these roots is

$$P_{b,\Omega}(z) = (z - b) \prod_{k=1}^{K} \left(z^2 - 2\sqrt{b}\cos(\omega_k) z + b \right). \tag{2.5}$$

Writing

$$P_{b,\Omega}(z) = z^m - \sum_{\ell=1}^{m} a_\ell(b,\Omega) z^{m-\ell} \tag{2.6}$$

defines the coefficient map $A : (b, \Omega) \mapsto a(b, \Omega)$.

Proposition 2.2. *For fixed (b, Ω), the rigid Leslie coefficients in* (2.6) *are unique. A rigid nonnegative Leslie realization exists if and only if*

$$a_\ell(b,\Omega) \geq 0, \qquad 1 \leq \ell \leq m. \tag{2.7}$$

Proof. The roots in (2.4) determine the monic polynomial (2.5). Comparison with (2.2) determines every coefficient. The companion matrix is entrywise nonnegative exactly when its first-row coefficients satisfy (2.7). □

Definition 2.3. The rigid admissible phase set is

$$\mathcal{A}_{b,K} = \left\{\Omega \in \mathbb{T}^K : a_\ell(b,\Omega) \geq 0 \text{ for all } \ell\right\}. \tag{2.8}$$

Remark 2.4. The boundary of (2.8) is contained in the union of the coefficient zero sets

$$\partial\mathcal{A}_{b,K} \subseteq \bigcup_{\ell=1}^{m} \{\Omega : a_\ell(b,\Omega) = 0\}. \tag{2.9}$$

Equality need not hold without excluding redundant inequalities and singular intersections.

3. Windowed cocycle and accumulated phase

Let $W_0, W_1, \ldots$ be consecutive windows in logarithmic scale. Associate a rigid Leslie matrix

$$L_j = L(a_j), \qquad a_j = a(b_j,\Omega_j) \in \mathbb{R}_+^m. \tag{3.1}$$

The non-autonomous evolution is

$$v_{j+1} = L_j v_j, \tag{3.2}$$

with cocycle product

$$\Phi(J,0) = L_{J-1}\cdots L_1 L_0. \tag{3.3}$$

Remark 3.1. A moving sequence L_j has no mandatory spectral periodicity. This does not imply that every chosen trajectory is nonperiodic. Periodicity may reappear if the coefficients or phases are selected periodically.

If a locally matched branch has phase $\omega_{j,k}$ in window j, its formal accumulated phase is

$$\Theta_k(J) = \sum_{j=0}^{J-1} \omega_{j,k}. \tag{3.4}$$

The definition requires mode matching and phase unwrapping between adjacent windows. Without those operations, the unordered roots of successive polynomials do not determine a unique trajectory $k \mapsto \omega_{j,k}$.

4. Exact coefficient coordinates for the rigid critical spectrum

The general inverse spectral problem for Leslie matrices has an established characterization in the literature [1]. The results in this section concern the special rigid spectrum in (2.4) and its use as a moving-window coordinate system.

Put $r = \sqrt{b}$ and define

$$S_\Omega(w) = \prod_{k=1}^{K} \left(w^2 - 2\cos(\omega_k)w + 1\right) = \sum_{j=0}^{2K} d_j w^{2K-j}. \tag{4.1}$$

The polynomial is reciprocal, and hence

$$d_{2K-j} = d_j, \qquad d_0 = d_{2K} = 1. \tag{4.2}$$

Moreover,

$$P_{b,\Omega}(z) = (z - r^2) r^{2K} S_\Omega(z/r). \tag{4.3}$$

Proposition 4.1. *The coefficients in* (2.6) *satisfy*

$$a_\ell = r^\ell \left(r d_{\ell-1} - d_\ell\right), \qquad 1 \le \ell \le 2K, \tag{4.4}$$

$$a_m = r^{m+1} = b^{K+1}. \tag{4.5}$$

For $1 \le j \le K$*, the paired coefficients are*

$$a_j = r^j \left(r d_{j-1} - d_j\right), \tag{4.6}$$

$$a_{m-j} = r^{m-j} \left(r d_j - d_{j-1}\right). \tag{4.7}$$

Proof. Expand (4.3) and compare the coefficient of $z^{m-\ell}$ with (2.6). Equation (4.7) follows from (4.2). □

Corollary 4.2. *A rigid nonnegative Leslie realization exists if and only if*

$$d_j > 0 \quad \textit{and} \quad \frac{1}{r} \le \frac{d_j}{d_{j-1}} \le r, \qquad 1 \le j \le K. \tag{4.8}$$

Strict inequalities are equivalent to strict positivity of all paired coefficients.

Proof. Equations (4.6) and (4.7) give the two inequalities. If one of d_{j-1} and d_j vanished while both paired inequalities held, the other would also vanish. Propagation back to $d_0 = 1$ yields a contradiction. Thus feasibility forces positivity of all d_j for $0 \le j \le K$. □

Definition 4.3. On the region where $d_1, \dots, d_K > 0$, define

$$\eta_j(\Omega) = \log \frac{d_j}{d_{j-1}}, \qquad 1 \le j \le K. \tag{4.9}$$

The rigid spectral strain is

$$\Sigma(\Omega) = \|\eta(\Omega)\|_\infty. \tag{4.10}$$

Corollary 4.4. *The positivity test becomes*

$$\|\eta(\Omega)\|_\infty \le \frac{1}{2} \log b. \tag{4.11}$$

The minimum compatible Perron value is

$$b_*(\Omega) = \exp\left(2\|\eta(\Omega)\|_\infty\right) \tag{4.12}$$

$$= \max_{1 \le j \le K} \left\{ \left(\frac{d_j}{d_{j-1}}\right)^2, \left(\frac{d_{j-1}}{d_j}\right)^2 \right\}. \tag{4.13}$$

If some $d_j \le 0$*, set* $b_*(\Omega) = +\infty$*.*

Remark 4.5. If $b > b_*(\Omega)$, the rigid Leslie coefficients are strictly positive. If $b = b_*(\Omega) < \infty$, at least one paired coefficient vanishes. The map b_* is generally nonsmooth where several coordinates attain the maximum in (4.12).

Proposition 4.6. *The rigid root-of-unity comb*

$$\omega_k = \frac{2\pi k}{2K+1}, \qquad 1 \le k \le K, \tag{4.14}$$

has $d_j = 1$ *for all* j*, and hence* $\Sigma(\Omega) = 0$*. Conversely,* $\Sigma(\Omega) = 0$ *implies the same multiset of phases, up to ordering and conjugation.*

Proof. For (4.14),

$$S_\Omega(w) = \frac{w^{2K+1} - 1}{w - 1} = 1 + w + \cdots + w^{2K}. \tag{4.15}$$

Conversely, $\eta = 0$ and $d_0 = 1$ imply $d_0 = \cdots = d_{2K} = 1$. The roots of (4.15) are the nontrivial $(2K+1)$-st roots of unity. □

5. Memory tails and the universal critical region

Normalize the coefficients by

$$x_\ell = \frac{a_\ell}{b^\ell}, \qquad X_q = \sum_{\ell=q}^{m} x_\ell. \tag{5.1}$$

Proposition 5.1. *The tails telescope to*

$$X_{j+1} = \frac{d_j}{r^j}, \qquad 0 \le j \le 2K. \tag{5.2}$$

Consequently,

$$X_{m-j} = b^{j-K} X_{j+1}, \qquad 0 \le j \le K, \tag{5.3}$$

$$x_m = X_m = b^{-K}. \tag{5.4}$$

Moreover, positivity is equivalent to

$$\frac{X_j}{b} \le X_{j+1} \le X_j, \qquad 1 \le j \le K. \tag{5.5}$$

Proof. Substitute (4.4) into (5.1); consecutive terms telescope. Equations (5.3)–(5.4) follow from (4.2), and (5.5) is (4.8) rewritten using (5.2). □

The reciprocal polynomial admits a Chebyshev reduction. Define

$$C_n(y) = 2T_n(y/2), \tag{5.6}$$

where T_n is the Chebyshev polynomial of the first kind, and set

$$H_\eta(y) = C_K(y) + d_1 C_{K-1}(y) + \cdots + d_{K-1} C_1(y) + d_K. \tag{5.7}$$

Then

$$S_\Omega(w) = w^K H_\eta(w + w^{-1}). \tag{5.8}$$

Definition 5.2. Let

$$\mathcal{R}_K = \left\{ \eta \in \mathbb{R}^K : H_\eta \text{ has } K \text{ simple roots in } (-2, 2) \right\}, \tag{5.9}$$

where $d_j = \exp(\eta_1 + \cdots + \eta_j)$.

Proposition 5.3. *In η-coordinates, the admissible rigid critical region is*

$$\mathcal{A}^\eta_{b,K} = \mathcal{R}_K \cap \left[-\frac{1}{2}\log b, \frac{1}{2}\log b \right]^K. \tag{5.10}$$

Thus $\mathcal{R}_K$ encodes the unit-circle root geometry, while the box encodes Leslie positivity.

Remark 5.4. Within the positive-d coordinates, a path can leave (5.10) either through a box face or through the boundary of $\mathcal{R}_K$. The latter occurs when roots collide or reach an endpoint. Since

$$H_\eta(2) = d_K + 2\sum_{j=0}^{K-1} d_j > 0, \tag{5.11}$$

a root cannot reach $+2$. Candidate spectral boundary equations are therefore

$$\operatorname{Disc} H_\eta = 0 \qquad \text{or} \qquad H_\eta(-2) = 0. \tag{5.12}$$

This describes the regular boundary mechanisms; intersections and higher degeneracies form singular strata.

6. Jacobian of the phase-to-strain map

Assume distinct phases $0 < \omega_k < \pi$. Let the phases be ordered to remove permutation ambiguity.

Proposition 6.1. *The coefficient Jacobian satisfies*

$$\left|\det \frac{\partial(d_1, \dots, d_K)}{\partial(\omega_1, \dots, \omega_K)}\right| = 2^{K(K+1)/2} \prod_{k=1}^{K} \sin \omega_k \prod_{i<j} |\cos \omega_i - \cos \omega_j|. \tag{6.1}$$

Consequently,

$$\left|\det \frac{\partial \eta}{\partial \Omega}\right| = \frac{2^{K(K+1)/2} \prod_{k} \sin \omega_k \prod_{i<j} |\cos \omega_i - \cos \omega_j|}{\prod_{j=1}^{K} d_j}. \tag{6.2}$$

Proof. The first K reciprocal coefficients determine the monic polynomial with roots $2\cos\omega_k$ in the Chebyshev basis. The root-to-coefficient Jacobian is a Vandermonde product, the change $2\cos\omega_k$ contributes $2\sin\omega_k$ in absolute value, and the triangular changes between monomial, Chebyshev, and reciprocal coefficient coordinates give the stated power of 2. The map $d \mapsto \eta$ is triangular with determinant $(d_1 \cdots d_K)^{-1}$. □

Corollary 6.2. *The same determinant can be written as*

$$\left|\det \frac{\partial \eta}{\partial \Omega}\right| = \frac{\sqrt{|\mathrm{Disc}\, H_\eta \, H_\eta(2) H_\eta(-2)|}}{d_1 d_2 \cdots d_K}. \tag{6.3}$$

Hence the phase-to-strain map is locally invertible away from phase collisions and the endpoints 0 *and* π*.*

Remark 6.3. At regular points, a moving rigid Leslie model can be transported using K variables η, plus b when the Perron value is allowed to vary. The full coefficient vector is reconstructed from (4.6), (4.7), and (4.5).

7. Exact viability conditions in strain coordinates

For fixed b, the box criterion (4.11) gives the exact first-order viability conditions. On an upper active face,

$$\eta_j = \frac{1}{2}\log b \quad \Longrightarrow \quad \dot\eta_j \le 0, \tag{7.1}$$

and on a lower active face,

$$\eta_j = -\frac{1}{2}\log b \quad \Longrightarrow \quad \dot\eta_j \ge 0. \tag{7.2}$$

Since $\dot\eta = D_\Omega \eta \, \dot\Omega$, both cases are summarized by

$$\mathrm{sgn}(\eta_j) \sum_{k=1}^{K} \frac{\partial \eta_j}{\partial \omega_k} \dot\omega_k \le 0 \tag{7.3}$$

on an active face.

If $b = b(t)$ varies and

$$\eta_j = s_j \frac{1}{2}\log b, \qquad s_j \in \{-1, 1\}, \tag{7.4}$$

then viability requires

$$s_j \dot\eta_j \le \frac{1}{2}\frac{\dot b}{b}. \tag{7.5}$$

For the minimum-Perron choice $b = b_*(\Omega)$ and a unique active index j_*,

$$\frac{\dot{b}_*}{b_*} = 2\,\mathrm{sgn}(\eta_{j_*})\dot{\eta}_{j_*}. \tag{7.6}$$

Remark 7.1. When several coordinates attain $\|\eta\|_\infty$, b_* is not differentiable in general. Directional derivatives or a differential inclusion must replace (7.6).

On a fixed upper face with tangential first-order motion, the second-order viability condition is

$$\nabla\eta_j \cdot \ddot{\Omega} + \dot{\Omega}^T D^2\eta_j \dot{\Omega} \le 0. \tag{7.7}$$

The inequality reverses on a lower face.

8. Exact modal transport and internally generated amplitudes

Assume that every rigid polynomial has distinct roots. For a root λ, define the right eigenvector

$$v(\lambda) = (\lambda^{m-1}, \lambda^{m-2}, \dots, \lambda, 1)^T. \tag{8.1}$$

Let

$$V_j = [v(\lambda_{j,1}) \ \cdots \ v(\lambda_{j,m})], \qquad \Lambda_j = \mathrm{diag}(\lambda_{j,1}, \dots, \lambda_{j,m}). \tag{8.2}$$

If L_j acts for h_j steps and $y_j = V_j c_j$, then

$$y_{j+1} = V_j \Lambda_j^{h_j} c_j, \tag{8.3}$$

$$c_{j+1} = T_j \Lambda_j^{h_j} c_j, \qquad T_j = V_{j+1}^{-1} V_j. \tag{8.4}$$

Proposition 8.1. *The entries of the basis-change matrix are*

$$(T_j)_{pq} = \prod_{\substack{s=1 \\ s\neq p}}^{m} \frac{\lambda_{j,q} - \lambda_{j+1,s}}{\lambda_{j+1,p} - \lambda_{j+1,s}}. \tag{8.5}$$

Proof. The p-th coordinate in the new Vandermonde basis is the cardinal Lagrange polynomial for the nodes $\lambda_{j+1,s}$, evaluated at the old node $\lambda_{j,q}$. Inverse Vandermonde matrices and Lagrange interpolation are equivalent descriptions of this change of basis. □

For a small root displacement

$$\lambda_{j+1,q} = \lambda_{j,q} + \delta\lambda_{j,q}, \tag{8.6}$$

one obtains, for $p \neq q$,

$$(T_j)_{pq} = -\delta\lambda_{j,q} \frac{P_j'(\lambda_{j,q})}{(\lambda_{j,q} - \lambda_{j,p}) P_j'(\lambda_{j,p})} + O(\|\delta\lambda_j\|^2). \tag{8.7}$$

Thus modal mixing becomes large near collisions, consistently with the vanishing of the discriminant.

Proposition 8.2. *The exact modal volume identity is*

$$\det T_j = \frac{\det V_j}{\det V_{j+1}}, \tag{8.8}$$

$$|\det T_j|^2 = \frac{|\operatorname{Disc} P_j|}{|\operatorname{Disc} P_{j+1}|}. \tag{8.9}$$

Proof. The first identity follows from $T_j = V_{j+1}^{-1} V_j$. The square of the absolute Vandermonde determinant equals the absolute discriminant of a monic polynomial with simple roots. □

Iterating (8.4) gives

$$c_J = T_{J-1}\Lambda_{J-1}^{h_{J-1}} \cdots T_0 \Lambda_0^{h_0} c_0, \tag{8.10}$$

and hence

$$y_J = V_J T_{J-1}\Lambda_{J-1}^{h_{J-1}} \cdots T_0 \Lambda_0^{h_0} c_0. \tag{8.11}$$

For the first-coordinate observable,

$$U_J = e_1^T V_J T_{J-1}\Lambda_{J-1}^{h_{J-1}} \cdots T_0 \Lambda_0^{h_0} c_0. \tag{8.12}$$

Warning 8.3. Equations (8.10)–(8.12) eliminate independent amplitude fitting only under the homogeneous state-handoff model (8.3). If each data window is reinitialized, forced, normalized independently, or connected through an additional observation operator, new parameters re-enter and must be counted explicitly.

9. Canonical boundary realization and the remaining law

Whenever $b_*(\Omega) < \infty$, define the canonical minimum-Perron realization

$$L_*(\Omega) = L\big(a(b_*(\Omega), \Omega)\big). \tag{9.1}$$

This realization lies on the boundary of the nonnegative cone. At regular points with unique active strain coordinate, the maps

$$\Omega \longleftrightarrow \eta \longmapsto b_*(\Omega) \longmapsto L_*(\Omega) \tag{9.2}$$

are locally determined, with the first arrow locally invertible.

Remark 9.1. The construction (9.1) is canonical but only piecewise smooth because of the maximum norm in b_*. It is also numerically sensitive near the spectral boundary (5.12).

The static coordinate chain (9.2) does not by itself determine how a future window is reached. The next sections derive an exact overlap law directly from the scalar sequence. For fixed b, the special reciprocal structure turns the admissible coefficient differences into a linear problem. At the first generic stride admitting motion, the entire update is reduced to one scalar innovation.

10. The fixed-Perron reciprocal affine hull

The critical-circle condition has a useful linear precursor. Fix $b > 1$, put $m = 2K + 1$, and for a coefficient vector $a = (a_1, \dots, a_m)$ define the synthetic-division coefficients

$$q_0(a; b) = 1, \tag{10.1}$$

$$q_j(a; b) = b^j - \sum_{\ell=1}^{j} a_\ell b^{j-\ell}, \qquad 1 \le j \le 2K. \tag{10.2}$$

Then

$$P(z; a) = (z - b)\left(z^{2K} + q_1 z^{2K-1} + \cdots + q_{2K}\right) + P(b; a). \tag{10.3}$$

Definition 10.1. The fixed-b reciprocal–Perron affine hull is the set

$$\mathcal{C}_{b,K}^{\mathrm{rec}} = \left\{ a \in \mathbb{R}^m : P(b; a) = 0, \quad q_{2K-j}(a; b) = b^{K-j} q_j(a; b),\ 0 \le j \le K - 1 \right\}. \tag{10.4}$$

Proposition 10.2. *The set* (10.4) *is an affine subspace of dimension K. Equivalently, there is a rank-$(K + 1)$ matrix R_b and a vector c_b such that*

$$\mathcal{C}_{b,K}^{\mathrm{rec}} = \{a : R_b a = c_b\}. \tag{10.5}$$

One explicit choice of the $K+1$ affine rows is

$$\sum_{\ell=1}^{m} b^{m-\ell} a_\ell = b^m, \tag{10.6}$$

$$\sum_{\ell=1}^{2K-j} b^{2K-j-\ell} a_\ell - \sum_{\ell=1}^{j} b^{K-\ell} a_\ell = b^{2K-j} - b^K, \qquad 0 \le j \le K-1. \tag{10.7}$$

Every rigid critical Leslie coefficient vector with Perron value b belongs to this affine subspace.

Proof. All equations in (10.4) are affine linear in a. The K quantities $q_1, \dots, q_K$ may be chosen freely. The reciprocal identities determine $q_{K+1}, \dots, q_{2K}$, after which

$$a_j = b q_{j-1} - q_j, \qquad 1 \le j \le 2K, \tag{10.8}$$

and $P(b; a) = 0$ determines

$$a_m = b q_{2K} = b^{K+1}. \tag{10.9}$$

Hence the affine dimension is K. For a rigid critical spectrum, $q_j = r^j d_j$ with $r = \sqrt{b}$, and the palindromy $d_{2K-j} = d_j$ gives exactly the reciprocal equations in (10.4). □

Remark 10.3. Membership in $\mathcal{C}^{\mathrm{rec}}_{b,K}$ is necessary but not sufficient for all non-Perron roots to lie on $|z| = \sqrt{b}$. Put

$$d_j = \frac{q_j}{b^{j/2}}. \tag{10.10}$$

The reciprocal identities give $d_{2K-j} = d_j$. Full criticality additionally requires the associated polynomial H in (5.7) to have all K roots in $[-2, 2]$, together with the Leslie nonnegativity inequalities.

11. Overlapping windows and exact mobility rigidity

Let (u_t) be any scalar sequence. In the intended arithmetic application one may take $u_t = M(B_t)$ or a predeclared normalization of those values. Define the recurrence row

$$h_t = (u_{t-1}, u_{t-2}, \dots, u_{t-m}) \in \mathbb{R}^m. \tag{11.1}$$

The order-m recurrence window beginning at n consists of the m equations

$$h_t a^{(n)} = u_t, \qquad n + m \le t \le n + 2m - 1. \tag{11.2}$$

When the corresponding square data matrix is nonsingular, these equations determine $a^{(n)}$ uniquely.

Let a second window begin at $n+s$, with $1 \le s \le m$. The two recurrence windows share exactly $m - s$ equations. Let $O_{n,s}$ be the matrix whose rows are the common h_t:

$$O_{n,s} = \begin{pmatrix} h_{n+s+m} \\ \vdots \\ h_{n+2m-1} \end{pmatrix} \in \mathbb{R}^{(m-s)\times m}, \tag{11.3}$$

with the empty-matrix convention when $s = m$.

Theorem 11.1 (Exact overlap constraint)**.** *Assume that $a^{(n)}$ and $a^{(n+s)}$ both fit their respective windows and both belong to $\mathcal{C}^{\mathrm{rec}}_{b,K}$ for the same fixed b. Then*

$$\delta a_{n,s} := a^{(n+s)} - a^{(n)} \in \ker O_{n,s} \cap \ker R_b. \tag{11.4}$$

Equivalently, with

$$Z_{n,s} = \begin{pmatrix} O_{n,s} \\ R_b \end{pmatrix}, \tag{11.5}$$

one has

$$Z_{n,s}\delta a_{n,s} = 0. \tag{11.6}$$

Proof. Every common recurrence equation is satisfied by both coefficient vectors, so subtraction gives $O_{n,s}\delta a_{n,s} = 0$. Since both vectors satisfy the same affine equations $R_b a = c_b$, subtraction also gives $R_b \delta a_{n,s} = 0$. □

Corollary 11.2 (Generic mobility threshold)**.** *If the rows of $O_{n,s}$ and R_b are in generic relative position, then*

$$\dim\ker Z_{n,s} = \max\{0, s-K-1\}. \tag{11.7}$$

Hence:

$$s \le K+1 \implies \textit{the old coefficient vector is locally isolated}, \tag{11.8}$$

$$s = K+2 \implies \textit{one generic coefficient direction survives}, \tag{11.9}$$

$$s > K+2 \implies s-K-1 \textit{ generic directions survive}. \tag{11.10}$$

Proof. The common-equation matrix contributes $m-s$ generic independent rows and R_b contributes $K+1$ independent rows. Since $m = 2K+1$, the generic nullity is

$$m - \min\{m, m-s+K+1\} = \max\{0, s-K-1\}. \tag{11.11}$$

□

Definition 11.3. For fixed b, define the critical stride

$$s_* = K+2. \tag{11.12}$$

It is the first stride at which a nonzero fixed-b reciprocal–Perron motion exists generically.

Remark 11.4. For $s \le K+1$, nontrivial motion is possible only when $Z_{n,s}$ loses rank. Thus exceptional motion is detected by the vanishing of maximal minors of $Z_{n,s}$. In particular, any single Hankel-type minor obtained from the fixed identity $a_m = b^{K+1}$ is only one necessary degeneracy condition, not a sufficient condition for full critical motion.

Remark 11.5. If b is allowed to vary, the regular rigid critical manifold has dimension $K+1$. A transversality count gives local intersection dimension

$$\max\{0, s-K\}, \tag{11.13}$$

so the first generic one-dimensional mobility occurs at $s = K+1$. Unlike the fixed-b result, this statement is local and nonlinear because coefficient differences for distinct Perron values do not lie in one fixed affine subspace.

12. Scalar innovation at the critical stride

Set $s = s_* = K+2$ and assume

$$\operatorname{rank} Z_{n,s_*} = m-1. \tag{12.1}$$

Choose a nonzero vector v_n spanning the one-dimensional nullspace,

$$\ker Z_{n,s_*} = \operatorname{span}\{v_n\}. \tag{12.2}$$

Theorem 12.1 (Scalarization of the window update)**.** *Under* (12.1)*, every next fixed-b reciprocal–Perron coefficient vector that agrees with the old window on all common equations has the form*

$$a^{(n+s_*)} = a^{(n)} + \tau_n v_n \tag{12.3}$$

for a unique scalar τ_n.

Proof. The difference belongs to the one-dimensional nullspace in (12.2). □

The next window contributes $s_* = K + 2$ genuinely new recurrence equations. The first occurs at

$$t_0 = n + 2m. \tag{12.4}$$

Its recurrence row h_{t_0} uses only values already known before u_{t_0} is observed.

Theorem 12.2 (One innovation sample determines the next Leslie)**.** *Assume*

$$h_{t_0} v_n \neq 0. \tag{12.5}$$

Then the first new scalar value determines the update parameter exactly:

$$\boxed{\tau_n = \frac{u_{t_0} - h_{t_0} a^{(n)}}{h_{t_0} v_n}.} \tag{12.6}$$

Consequently the entire next coefficient vector is fixed by (12.3)*.*

Proof. Substitute (12.3) into the recurrence equation at t_0 and solve the resulting scalar equation for τ_n. □

Corollary 12.3 (Causal $K+1$-value forecast)**.** *After observing only u_{t_0}, define $a^+ = a^{(n)} + \tau_n v_n$ using* (12.6)*. Then recursively define*

$$\widehat{u}_t = \sum_{\ell=1}^{m} a_\ell^+ \widehat{u}_{t-\ell}, \qquad t_0 + 1 \le t \le t_0 + K + 1, \tag{12.7}$$

using the observed historical values for indices below t_0 and earlier forecasts above t_0. If the next window is exactly generated by a fixed-b rigid critical Leslie coefficient vector, then

$$\widehat{u}_t = u_t, \qquad t_0 + 1 \le t \le t_0 + K + 1. \tag{12.8}$$

Proof. The first new observation determines the exact next coefficient vector. The remaining $K+1$ new equations of that window use the same coefficient vector. Induction on t in (12.7) gives the result. □

Proposition 12.4 (Rank-one block diagnostic)**.** *Let G_n contain the $K+2$ new recurrence rows of the next window, evaluated from the observed sequence, and let y_n^+ be their right-hand side. Define*

$$e_n = y_n^+ - G_n a^{(n)}, \qquad g_n = G_n v_n. \tag{12.9}$$

Then exact critical-stride transport implies

$$e_n = \tau_n g_n. \tag{12.10}$$

Equivalently,

$$(e_n)_i (g_n)_j - (e_n)_j (g_n)_i = 0 \qquad \text{for all } i, j. \tag{12.11}$$

Thus a block of $K+2$ new equations carries only one scalar innovation after the old state and overlap direction are fixed.

13. Admissible innovation interval and boundary locking

The scalar parameter in (12.3) cannot take arbitrary values if nonnegativity and the critical-circle root condition are retained.

Definition 13.1. For a current nonnegative coefficient vector a and direction v, define the interval of coefficient positivity

$$I_{\mathrm{pos}}(a, v) = \left[\max_{v_\ell > 0} \left(-\frac{a_\ell}{v_\ell} \right), \ \min_{v_\ell < 0} \left(-\frac{a_\ell}{v_\ell} \right) \right], \tag{13.1}$$

with the usual $-\infty$ and $+\infty$ conventions when one of the indexing sets is empty.

For $a(\tau) = a + \tau v$, define $q_j(\tau)$ by (10.2), $d_j(\tau) = q_j(\tau)/b^{j/2}$, and the corresponding reciprocal reduction H_τ. Let I_{spec} be the connected component containing 0 of the set of τ for which H_τ has K simple roots in $(-2, 2)$. At regular points, an endpoint of this component can occur only when

$$\operatorname{Disc} H_\tau = 0 \qquad \text{or} \qquad H_\tau(-2) = 0. \tag{13.2}$$

Define the full local innovation interval

$$I_{\mathrm{adm}} = I_{\mathrm{pos}} \cap I_{\mathrm{spec}}. \tag{13.3}$$

Corollary 13.2 (Pre-observation admissible range)**.** *Before u_{t_0} is observed, the moving rigid Leslie hypothesis requires*

$$\boxed{u_{t_0} \in h_{t_0} a^{(n)} + (h_{t_0} v_n) I_{\mathrm{adm}}.} \tag{13.4}$$

A value outside this interval falsifies the local fixed-b critical transition before any later value of the new block is used.

Proposition 13.3 (Boundary locking criterion)**.** *Let*

$$\mathcal{Z}(a) = \{\ell : a_\ell = 0\} \tag{13.5}$$

be the active coefficient faces. A nonzero one-sided coefficient motion $a+\tau v$ preserving nonnegativity exists locally only if the numbers v_ℓ, $\ell \in \mathcal{Z}(a)$, do not contain both positive and negative values. If both signs occur, then $\tau = 0$ is the only sufficiently small feasible update.

Proof. For each active coordinate the inequality $a_\ell + \tau v_\ell \geq 0$ reduces to $\tau v_\ell \geq 0$. A common nonzero sign of τ exists exactly when the active v_ℓ do not impose conflicting signs. □

Remark 13.4. The canonical minimum-Perron realization $L_*(\Omega)$ always has at least one active coefficient face. It is therefore dynamically more restrictive than an interior choice $b > b_*(\Omega)$. A canonical boundary trajectory may lock whenever the critical-stride direction intersects active faces with incompatible signs.

14. Exact scalar-driven strain transport

For the update $a^+ = a + \tau v$, define

$$\psi_j(v; b) = \sum_{\ell=1}^{j} v_\ell b^{j-\ell}, \qquad \psi_0 = 0. \tag{14.1}$$

Equation (10.2) gives

$$q_j(a + \tau v; b) = q_j(a; b) - \tau \psi_j(v; b). \tag{14.2}$$

Theorem 14.1 (Exact strain update)**.** *Whenever all logarithms are defined,*

$$\boxed{\eta_j^+ - \eta_j = \log\left(1 - \tau \frac{\psi_j}{q_j}\right) - \log\left(1 - \tau \frac{\psi_{j-1}}{q_{j-1}}\right), \qquad 1 \leq j \leq K.} \tag{14.3}$$

Hence, at the critical stride, all K strain coordinates are driven by the same scalar innovation τ.

Proof. Use $d_j = q_j/b^{j/2}$ in the definition $\eta_j = \log(d_j/d_{j-1})$ and substitute (14.2). □

Differentiating with respect to the scalar innovation gives the exact line-flow equation

$$\frac{d\eta_j}{d\tau} = -\frac{\psi_j}{q_j - \tau\psi_j} + \frac{\psi_{j-1}}{q_{j-1} - \tau\psi_{j-1}}. \tag{14.4}$$

For small τ,

$$\eta_j^+ - \eta_j = -\tau\left(\frac{\psi_j}{q_j} - \frac{\psi_{j-1}}{q_{j-1}}\right) + O(\tau^2). \tag{14.5}$$

Thus the K-dimensional strain increment is generically confined to one first-order direction at the critical stride.

15. Frequency-free determinant identities

The reciprocal–Perron test can be imposed directly on a data window without first estimating any phases. Let

$$\mathsf{A}_n = \begin{pmatrix} h_{n+m} \\ \vdots \\ h_{n+2m-1} \end{pmatrix}, \qquad \mathsf{g}_n = \begin{pmatrix} u_{n+m} \\ \vdots \\ u_{n+2m-1} \end{pmatrix}. \tag{15.1}$$

Write the affine reciprocal constraints in (10.5) row by row as

$$\rho_\alpha(b)^T a = \kappa_\alpha(b), \qquad 0 \le \alpha \le K. \tag{15.2}$$

For each row define

$$\Delta_{\alpha,n}(b) = \det \begin{pmatrix} \mathsf{A}_n & \mathsf{g}_n \\ \rho_\alpha(b)^T & \kappa_\alpha(b) \end{pmatrix}, \tag{15.3}$$

Proposition 15.1. *If* $\det \mathsf{A}_n \neq 0$*, then the unique order-m recurrence coefficient vector of the window belongs to* $\mathcal{C}^{\mathrm{rec}}_{b,K}$ *if and only if*

$$\Delta_{\alpha,n}(b) = 0, \qquad 0 \le \alpha \le K. \tag{15.4}$$

Full rigid criticality is obtained by adding the nonnegativity and root-location tests of Sections 4–5.

Proof. Since $a = \mathsf{A}_n^{-1}\mathsf{g}_n$, the Schur-complement identity gives

$$\Delta_{\alpha,n}(b) = \det(\mathsf{A}_n)\left(\kappa_\alpha(b) - \rho_\alpha(b)^T \mathsf{A}_n^{-1}\mathsf{g}_n\right). \tag{15.5}$$

Thus the determinants vanish exactly when all affine reciprocal–Perron constraints hold. □

Remark 15.2. If b is not fixed in advance, the functions $\Delta_{\alpha,n}(b)$ are explicit polynomials in b whose coefficients depend only on the data window. A necessary condition for a rigid critical realization with some $b > 1$ is that these $K+1$ polynomials possess a common real root $b > 1$. This can be tested without frequencies by polynomial gcd or resultant/subresultant elimination, followed by the root-location and positivity conditions.

16. Closed form for one mode pair

For $K = 1$ one has $m = 3$, and the entire construction becomes explicit. Write $\alpha = a_1$. The fixed-b reciprocal–Perron constraints give

$$\boxed{a_1 = \alpha, \qquad a_2 = b^2 - b - b\alpha, \qquad a_3 = b^2.} \tag{16.1}$$

The quotient polynomial is

$$Q(z) = z^2 + (b - \alpha)z + b. \tag{16.2}$$

Its two roots lie on $|z| = \sqrt{b}$, and the Leslie coefficients are nonnegative, if and only if

$$\boxed{\max\{0, b - 2\sqrt{b}\} \le \alpha \le b - 1.} \tag{16.3}$$

Endpoint equalities include coefficient or spectral degeneracies; strict interior conditions give the corresponding nondegenerate case.

Define

$$X_t(b) = u_{t-1} - bu_{t-2}, \tag{16.4}$$

$$Y_t(b) = u_t - (b^2 - b)u_{t-2} - b^2 u_{t-3}. \tag{16.5}$$

Then the critical recurrence is simply

$$Y_t(b) = \alpha X_t(b). \tag{16.6}$$

Thus, if (X_t, X_{t+1}, X_{t+2}) is not identically zero, three consecutive recurrence equations admit a common fixed-b reciprocal coefficient vector if and only if all pairwise minors vanish:

$$X_i Y_j - X_j Y_i = 0, \qquad t \le i < j \le t+2. \tag{16.7}$$

If $X_t = X_{t+1} = X_{t+2} = 0$, existence instead requires $Y_t = Y_{t+1} = Y_{t+2} = 0$. Full rigid criticality additionally requires the inferred α to satisfy (16.3).

For $K = 1$, the critical stride is $s_* = 3$ and the fixed-b affine direction is

$$v = (1, -b, 0)^T. \tag{16.8}$$

Hence

$$\alpha^+ = \alpha + \tau, \qquad a_2^+ = a_2 - b\tau, \qquad a_3^+ = b^2. \tag{16.9}$$

The first new recurrence equation gives

$$\tau = \frac{u_{t_0} - h_{t_0} a}{X_{t_0}(b)}, \tag{16.10}$$

provided $X_{t_0}(b) \neq 0$, and the next two values are then fixed recursively. The exact admissible scalar interval is

$$\boxed{\max\{0, b - 2\sqrt{b}\} - \alpha \le \tau \le b - 1 - \alpha.} \tag{16.11}$$

This gives a completely frequency-free three-step falsification and two-value forecasting rule for the one-pair model.

17. Irreducible scalar forcing and the endpoint of the Leslie geometry

Proposition 17.1. *At a regular interior fixed-b point and the critical stride $s_* = K + 2$, the Leslie geometry cannot determine τ_n from the old window and the common equations alone. In a neighborhood of $\tau = 0$, distinct values of τ produce distinct reciprocal–Perron coefficient vectors satisfying all those old and common constraints. Full criticality persists for a nontrivial interval whenever the current point lies in the interior of the fixed-b critical region.*

Proof. The rank assumption (12.1) leaves exactly the affine line (12.3). Within the reciprocal affine hull, coefficient positivity and the condition that the reduced polynomial have simple roots in $(-2, 2)$ are open. Hence sufficiently small values of τ remain admissible. □

Remark 17.2. This identifies the exact limit of the moving-Leslie mechanism. The geometry compresses a block of $K + 2$ genuinely new recurrence equations to one scalar innovation and then transports all K strain coordinates through (14.3). It does not supply a law for that scalar innovation. A zero-latency forecast of the first new value therefore requires an additional arithmetic statement about the sequence τ_n. Any such law is external to Leslie geometry and is the remaining place where genuinely Möbius-specific information would have to enter.

18. Exact first-order spectral transport

Let $a(t)$ be a differentiable coefficient path and define

$$P(z, t) = z^m - \sum_{\ell=1}^{m} a_\ell(t) z^{m-\ell}. \tag{18.1}$$

Let $\lambda_k(t)$ be a simple root, so that

$$P(\lambda_k(t), t) = 0, \qquad P_z(\lambda_k(t), t) \neq 0. \tag{18.2}$$

Differentiability of simple eigenvalue branches is standard in matrix perturbation theory [3].

Proposition 18.1. *Under* (18.2)*,*

$$\dot{\lambda}_k = \frac{\displaystyle\sum_{\ell=1}^{m} \dot{a}_\ell \lambda_k^{m-\ell}}{P_z(\lambda_k, t)}. \tag{18.3}$$

Proof. Differentiating $P(\lambda_k(t), t) = 0$ gives

$$P_z(\lambda_k, t)\dot{\lambda}_k + P_t(\lambda_k, t) = 0. \tag{18.4}$$

Since

$$P_t(z, t) = -\sum_{\ell=1}^{m} \dot{a}_\ell z^{m-\ell}, \tag{18.5}$$

substitution into (18.4) yields (18.3). □

Suppose now that

$$\lambda_k(t) = \sqrt{b(t)}e^{i\omega_k(t)}. \tag{18.6}$$

Then

$$\frac{\dot{\lambda}_k}{\lambda_k} = \frac{\dot{b}}{2b} + i\dot{\omega}_k. \tag{18.7}$$

Define

$$J_{k\ell}(L) = \frac{\lambda_k^{m-\ell-1}}{P_z(\lambda_k)}. \tag{18.8}$$

Corollary 18.2. *For every simple prescribed branch,*

$$\dot{\omega}_k = \Im\left(\sum_{\ell=1}^{m} J_{k\ell}(L)\dot{a}_\ell\right), \tag{18.9}$$

$$\frac{\dot{b}}{2b} = \Re\left(\sum_{\ell=1}^{m} J_{k\ell}(L)\dot{a}_\ell\right). \tag{18.10}$$

Warning 18.3. Equation (18.10) must give the same scalar $\dot{b}/(2b)$ for every k. Thus the imaginary-part formula alone is not a complete admissibility test. The real-part compatibility equations are additional local constraints.

19. The rigid critical manifold

Let

$$\mathcal{M}_K = \Big\{a(b, \Omega) : b > 1,\ \Omega \in \mathbb{T}^K,\ a(b, \Omega) \geq 0\Big\} \subseteq \mathbb{R}^{2K+1}. \tag{19.1}$$

Proposition 19.1. *At a point where the coefficient map $(b, \Omega) \mapsto a(b, \Omega)$ has rank $K + 1$ and the prescribed roots are distinct, $\mathcal{M}_K$ is locally a smooth manifold of dimension $K+1$ and codimension K in $\mathbb{R}^{2K+1}$.*

Proof. The parameter space has dimension $K + 1$. Under the stated full-rank hypothesis, the constant-rank theorem gives a local image manifold of dimension $K + 1$. The ambient dimension is $2K + 1$, so the codimension is K. □

Consequently, locally there are K independent constraints

$$F_r(a) = 0, \qquad 1 \leq r \leq K, \tag{19.2}$$

and every differentiable path in the regular part satisfies

$$\nabla F_r(a) \cdot \dot{a} = 0, \qquad 1 \leq r \leq K. \tag{19.3}$$

Warning 19.2. The dimension statement is generic, not universal. Repeated roots, collisions of phases, loss of rank, and intersections with several active coefficient faces can create singular or lower-dimensional strata.

20. Positivity and the corrected spectral velocity cone

For $a \in \mathbb{R}^m_+$, the tangent cone of the nonnegative orthant is

$$T_a\mathbb{R}^m_+ = \{v \in \mathbb{R}^m : v_\ell \geq 0 \text{ whenever } a_\ell = 0\}. \tag{20.1}$$

Tangent cones generalize tangent spaces at nonsmooth boundary points of convex sets [4].

The velocity must satisfy three types of conditions simultaneously:

(1) orthant feasibility, $v \in T_a\mathbb{R}^m_+$;
(2) tangency to the rigid critical manifold, $DF(a)v = 0$;
(3) common radial motion across all prescribed root pairs.

Define the common-radial constraint operator by

$$\mathcal{R}_L v = (\Re(J_2.(L)v) - \Re(J_1.(L)v), \ldots, \Re(J_K.(L)v) - \Re(J_1.(L)v))^T. \tag{20.2}$$

At a regular point, define the admissible coefficient-velocity cone

$$\mathcal{K}(L) = T_a\mathbb{R}^m_+ \cap \ker DF(a) \cap \ker \mathcal{R}_L. \tag{20.3}$$

Definition 20.1. The rigid positive spectral velocity cone is

$$\mathcal{V}(L) = \{\Im J(L)v : v \in \mathcal{K}(L)\}. \tag{20.4}$$

Proposition 20.2. *Every differentiable rigid positive Leslie path with simple prescribed roots satisfies*

$$\dot{\Omega}(t) \in \mathcal{V}(L(t)). \tag{20.5}$$

Proof. The coefficient velocity lies in the orthant tangent cone, is tangent to the rigid critical manifold, and satisfies the common-radial identities. Equation (18.9) then places its phase velocity in the image (20.4). □

Remark 20.3. Condition (20.5) is first-order and necessary. At singular points or for finite window increments it is not sufficient for the existence of a global path.

21. Second-order transport

Proposition 21.1. *If $a(t)$ is twice differentiable and* (18.2) *holds, then*

$$\ddot{\lambda}_k = -\frac{P_{zz}\dot{\lambda}_k^2 + 2P_{zt}\dot{\lambda}_k + P_{tt}}{P_z}, \tag{21.1}$$

where all polynomial derivatives are evaluated at $(\lambda_k(t), t)$ and

$$P_{tt}(z,t) = -\sum_{\ell=1}^{m} \ddot{a}_\ell z^{m-\ell}. \tag{21.2}$$

Proof. Differentiate (18.4) once more and collect the chain-rule terms. Division by $P_z \neq 0$ gives (21.1). □

For the parameterization (18.6),

$$\frac{\ddot{\lambda}_k}{\lambda_k} = \frac{\ddot{b}}{2b} - \frac{\dot{b}^2}{4b^2} - \dot{\omega}_k^2 + i\left(\ddot{\omega}_k + \frac{\dot{b}}{b}\dot{\omega}_k\right). \tag{21.3}$$

Thus a smooth Leslie path constrains not only Ω and $\dot{\Omega}$, but also the angular acceleration after the radial terms are removed.

Definition 21.2. At a regular path point, the residual obtained by comparing the observed $\ddot{\Omega}$ with the imaginary part implied by (21.1) is called the local Leslie spectral-curvature residual.

Warning 21.3. A numerical curvature residual depends on the chosen interpolation of discrete windows and on the time scale assigned to the window index. It is not intrinsic until those choices are fixed before testing.

22. Discrete windows and finite increments

The observations are discrete, not a differentiable curve. Let

$$\Delta a_j = a_{j+1} - a_j, \qquad \Delta\Omega_j = \operatorname{unwrap}(\Omega_{j+1} - \Omega_j). \tag{22.1}$$

The differential model suggests the residual

$$r_j = \Delta\Omega_j - \Im J(L_j)\Delta a_j. \tag{22.2}$$

Warning 22.1. Equation (22.2) is only a first-order approximation. A valid finite-step test should also check that the segment or reconstructed path remains in the admissible set and should report an error scale of order $\|\Delta a_j\|^2$ away from singularities.

A stronger discrete feasibility test is to solve for a path $a_j(s)$, $0 \le s \le 1$, satisfying

$$a_j(0) = a_j, \qquad a_j(1) = a_{j+1}, \qquad a_j(s) \in \mathcal{M}_K \cap \mathbb{R}^m_+. \tag{22.3}$$

Existence of (22.3) is stronger than endpoint feasibility.

23. A genuine forecasting protocol

Experiment 23.1. Choose before examining the test windows:

(1) a fixed base b or a predeclared rule for b_j;
(2) a fixed number K of mode pairs and hence $m = 2K + 1$;
(3) consecutive, nonoverlapping or explicitly overlapping logarithmic windows;
(4) a frequency estimator and a deterministic mode-matching rule;
(5) a coefficient-transport law trained only on past windows;
(6) a rule for handling windows outside $\mathcal{A}_{b,K}$;
(7) a horizon and evaluation metric.

For each training endpoint J, infer $L_0, \dots, L_J$, predict $\widehat{a}_{J+1}$ without using W_{J+1}, project to the admissible rigid set if that projection was predeclared, compute $\widehat{\Omega}_{J+1}$ from the roots, and only then evaluate the prediction on W_{J+1}.

The information flow must be

$$M|_{W_0,\dots,W_J} \longrightarrow L_0, \dots, L_J \longrightarrow \widehat{L}_{J+1} \longrightarrow \widehat{\Omega}_{J+1} \longrightarrow \widehat{M}|_{W_{J+1}}. \tag{23.1}$$

Warning 23.2. The identities of Sections 18–21 do not themselves specify a forecasting law for $\dot{a}$ or $\ddot{a}$. A predictor requires a predeclared dynamical model for coefficient evolution. Merely estimating L_{J+1} from the future window would return to independent fitting.

24. Critical-stride forecasting protocol

Experiment 24.1. Fix K, $m = 2K + 1$, a Perron value $b > 1$, and the critical stride $s_* = K + 2$ before examining the test region. For each training window:

(1) solve the m recurrence equations (11.2) directly from the past scalar sequence;
(2) reject the window unless the resulting coefficients satisfy the fixed-b reciprocal determinants (15.4), nonnegativity, and the critical-circle root-location test;
(3) construct Z_{n,s_*} using only common equations and R_b;
(4) require $\operatorname{rank} Z_{n,s_*} = m - 1$ and compute its null direction v_n;

(5) compute the pre-observation interval (13.4);
(6) reveal only the first new scalar u_{t_0} and compute τ_n from (12.6);
(7) without recalibration, predict the remaining $K+1$ new values using (12.7);
(8) record the $K+1$ forecast errors, the scalar innovation τ_n, the strain update (14.3), and whether $\tau_n \in I_{\mathrm{adm}}$.

The information flow in this strongest fixed-b test is therefore

$$\text{past block} \longrightarrow (a^{(n)}, v_n, I_{\mathrm{adm}}) \longrightarrow u_{t_0} \longrightarrow \tau_n \longrightarrow \widehat{u}_{t_0+1}, \dots, \widehat{u}_{t_0+K+1}, \tag{24.1}$$

where u_{t_0} is the single newly revealed scalar value; no later test value is used to update the coefficients.

Remark 24.2. If the $K+1$-value forecasts succeed repeatedly, the remaining arithmetic object is the scalar innovation sequence (τ_n). A further predictive law for τ_n would upgrade the one-sample-delayed forecast to a genuine zero-latency block forecast. Leslie geometry alone does not provide such a law by Proposition 17.1.

25. Three nested models

Definition 25.1. The comparison should separate:

(1) free frequencies fitted independently in every window;
(2) rigid nonnegative Leslie matrices fitted independently in every window;
(3) one dynamically transported rigid nonnegative Leslie trajectory, forecast from past windows only.

Model B tests static realizability. Model C tests whether a low-complexity trajectory inside the constrained geometry has forecasting content. Model C should be compared not only with Models A and B, but also with persistence, autoregressive, phase-randomized, and support-preserving null models.

26. What the previous forty runs do and do not establish

Experiment 26.1. The reported forty runs varied the sampling base, training fraction, and requested number of modes. They established numerical out-of-sample prediction by free frequencies and static positive spectral completion with auxiliary eigenvalues for those forty configurations.

Proposition 26.2. *The forty configurations do not constitute a temporal Leslie trajectory and cannot test* (20.5) *or the forecast protocol* (23.1)*.*

Proof. A temporal test requires consecutively ordered windows under fixed or predeclared modelling choices. The forty runs index alternative parameter configurations rather than successive time windows of one fixed model. □

27. Verified corrections to the proposed formulation

Proposition 27.1. *The first derivative* (18.3)*, the angular and radial decompositions* (18.9)–(18.10)*, and the second derivative* (21.1) *are algebraically correct under the simple-root hypothesis.*

Remark 27.2. An independent finite-difference calculation was performed for a degree-five companion polynomial with randomly chosen coefficient velocities. The maximum discrepancy between the finite-difference root velocity and (18.3) was below 9×10^{-9} for a step of 10^{-7}. This is a numerical consistency check, not a substitute for Proposition 18.1.

Remark 27.3. Three corrections are essential:

(1) the boundary relation is the inclusion (2.9), not automatic equality;

(2) the spectral velocity cone must include tangency to the rigid manifold and common radial compatibility, as in (20.3);
(3) the dimension and codimension counts are generic and require a full-rank hypothesis.

28. Numerical obstruction for the rigid $K = 7$ core

The rigid feasibility criterion of Corollary 4.2 can be evaluated even when the logarithmic coordinates η_j do not exist. For $r = \sqrt{b}$ define

$$\delta_j(b, \Omega) = \min\left\{rd_{j-1} - d_j, rd_j - d_{j-1}\right\}, \qquad 1 \le j \le K, \tag{28.1}$$

and

$$\Delta_K(b, \Omega) = \min_{1\le j\le K} \delta_j(b, \Omega). \tag{28.2}$$

Because the factors multiplying the two quantities in (28.1) in the paired Leslie coefficients are positive, one has

(28.3)
$$\Delta_K(b, \Omega) \ge 0 \quad \Longleftrightarrow \quad P_{b,\Omega} \text{ is the characteristic polynomial of a nonnegative rigid Leslie matrix.}$$

Unlike $b_*(\Omega)$, the margin (28.2) remains defined when one or more d_j are nonpositive.

Experiment 28.1. A grid of 241 values of b in $[1.01, 1.25]$ was examined with $K = 7$ and the predeclared frequency estimator. The estimator returned all seven frequencies at eight grid points. For every one of these eight evaluable configurations,

$$\Delta_7(b, \Omega) < 0. \tag{28.4}$$

The largest observed margin was

$$\boxed{\Delta_{\max} = -1.228610846762} \qquad \text{at} \qquad \boxed{b = 1.178}. \tag{28.5}$$

Thus none of the eight evaluable critical spectra is the complete spectrum of a nonnegative Leslie matrix of rigid dimension 15.

Seven of the eight configurations had $d_1, \dots, d_7 > 0$ and therefore admitted the logarithmic coordinates η and the finite quantity $b_*(\Omega)$. In all seven cases $b_*(\Omega) > b$. The eighth configuration had all seven frequencies but at least one nonpositive d_j, so b_* was not finite although Δ_7 remained evaluable.

The index attaining the most negative paired margin had the empirical distribution

j_*	2	3	4	5	6
count	1	1	3	1	2.

(28.6)

In particular, $j_* = 1$ did not occur. The dominant obstruction in these eight cases is therefore not the first hemispheric condition alone; it occurs in higher symmetric combinations of the seven phases.

Remark 28.2. Experiment 28.1 is a statement about the eight grid points at which the full $K = 7$ core was available. The other 233 points are not evaluations of Δ_7 and must not be counted as rigidly feasible or infeasible. Likewise, a reported prefix statistic $\kappa = 0$ only excludes the tested angularly ordered prefixes; it does not exclude every possible subfamily of the seven frequencies.

Corollary 28.3. *For each configuration in Experiment 28.1 for which a nonnegative Leslie realization containing the same prescribed roots exists in a larger dimension, at least one auxiliary root is necessary.*

Proof. Equation (28.4) excludes degree-zero completion, i.e. the rigid characteristic polynomial itself. Any larger realization containing the same roots therefore has a nonconstant complementary factor. □

29. Minimal auxiliary completion as an exact linear feasibility problem

The negative rigid test does not determine how many auxiliary eigenvalues are necessary. That question has an exact finite-dimensional formulation that does not require re-estimating any frequency.

Let

$$C(z) = z^d + c_1 z^{d-1} + \cdots + c_d, \qquad d = 2K+1, \tag{29.1}$$

be a fixed rigid core polynomial $P_{b,\Omega}$, so $C(b) = 0$. For $q \geq 0$ let

$$R_q(z) = z^q + r_1 z^{q-1} + \cdots + r_q \tag{29.2}$$

be a real monic polynomial and write

$$C(z)R_q(z) = z^{d+q} + \sum_{k=1}^{d+q} p_k z^{d+q-k}. \tag{29.3}$$

Definition 29.1. The auxiliary completion index of the core C is

$$\nu(C) = \min\left\{q \geq 0 : \exists R_q \text{ monic real with } p_k \leq 0 \text{ for every } k \geq 1\right\}, \tag{29.4}$$

with $\nu(C) = +\infty$ if no such polynomial exists. For $C = P_{b,\Omega}$ we also write $\nu_b(\Omega) = \nu(C)$.

Proposition 29.2. *A degree-q auxiliary completion in Definition 29.1 exists if and only if there is a nonnegative Leslie companion matrix of dimension $d+q$ whose spectrum contains all roots of C and whose Perron eigenvalue is b.*

Proof. If $p_k \leq 0$, set $a_k = -p_k \geq 0$. Then (29.3) is

$$z^{d+q} - \sum_{k=1}^{d+q} a_k z^{d+q-k},$$

the characteristic polynomial of a nonnegative Leslie companion matrix. Since $C(b) = 0$, the product also vanishes at b. The vector

$$(b^{d+q-1}, b^{d+q-2}, \ldots, b, 1)^T$$

is strictly positive and is an eigenvector with eigenvalue b. The Collatz–Wielandt characterization therefore gives spectral radius b.

Conversely, if a nonnegative Leslie characteristic polynomial contains all roots of C, polynomial division by the monic factor C gives a real monic complementary polynomial R_q, and nonnegativity of the Leslie coefficients is exactly the condition $p_k \leq 0$. □

Set $c_0 = 1$ and $c_k = 0$ outside $0 \leq k \leq d$, and set $r_0 = 1$. Then

$$p_k = \sum_{j=0}^{q} r_j c_{k-j} = c_k + \sum_{j=1}^{q} c_{k-j} r_j. \tag{29.5}$$

Define the $(d+q) \times q$ Toeplitz matrix

$$(A_q)_{kj} = c_{k-j}, \qquad 1 \leq k \leq d+q, \quad 1 \leq j \leq q, \tag{29.6}$$

and the vector

$$h_q = (-c_1, \ldots, -c_d, 0, \ldots, 0)^T \in \mathbb{R}^{d+q}. \tag{29.7}$$

Theorem 29.3. *For fixed q, a degree-q auxiliary completion exists if and only if the linear system*

$$\boxed{A_q r \leq h_q} \tag{29.8}$$

is feasible for $r = (r_1, \dots, r_q)^T$. *Consequently,*

$$\boxed{\nu(C) = \min\{q \geq 0 : A_q r \leq h_q \text{ is feasible}\}.} \tag{29.9}$$

Proof. Equation (29.5) gives $p_k \leq 0$ exactly as the kth inequality in (29.8). Definition 29.1 then gives (29.9). □

Proposition 29.4 (Dual infeasibility certificate)**.** *Degree q is impossible if and only if there exists a vector $y \in \mathbb{R}^{d+q}$ satisfying*

$$\boxed{y \geq 0, \qquad A_q^T y = 0, \qquad h_q^T y < 0.} \tag{29.10}$$

Such a y is a Farkas certificate that no choice of q auxiliary roots can repair the rigid core.

Proof. This is Farkas' alternative applied to the system $A_q r \leq h_q$. □

Proposition 29.5. *If degree q is feasible, every degree $q' \geq q$ is feasible. Hence the feasible auxiliary degrees form either the empty set or the tail*

$$\{\nu(C), \nu(C)+1, \nu(C)+2, \dots\}. \tag{29.11}$$

Proof. If R_q is feasible, then zR_q is monic of degree $q+1$ and

$$C(z) z R_q(z) = z\, C(z) R_q(z).$$

Its nonleading coefficients are the previous nonpositive coefficients shifted by one place, together with a final zero. Iteration proves the claim. □

29.1. **Auxiliary defect profile.** For a quantitative profile before exact feasibility, define

$$E_q(C) = \min_{r \in \mathbb{R}^q,\ e \geq 0} \{e : A_q r - h_q \leq e\mathbf{1}\}. \tag{29.12}$$

Equivalently, E_q is the minimum possible largest positive coefficient of the completed product. To compare different b values, the same definition may be applied after the sign-preserving scaling $z = \sqrt{b}\, w$, which replaces c_k by $c_k / b^{k/2}$.

Proposition 29.6. *The auxiliary defect profile satisfies*

$$E_q(C) \geq 0, \qquad E_{q+1}(C) \leq E_q(C), \qquad E_q(C) = 0 \iff q \geq \nu(C). \tag{29.13}$$

Thus $q \mapsto E_q(C)$ records how the rigid obstruction is removed as auxiliary degrees are introduced, while $\nu(C)$ is its first zero.

Proof. Nonnegativity is built into (29.12). The construction $R_q \mapsto zR_q$ used in Proposition 29.5 preserves the coefficient violation and proves monotonicity. Finally, $E_q = 0$ is exactly feasibility of (29.8). □

29.2. **Exact one-auxiliary test.** Because the rigid core has

$$c_d = -b^{K+1} < 0, \tag{29.14}$$

a degree-one multiplier $R_1(z) = z + t$ must satisfy $t \geq 0$. Its remaining conditions are

$$c_1 + t \leq 0, \qquad c_k + t c_{k-1} \leq 0, \quad 2 \leq k \leq d. \tag{29.15}$$

Hence $\nu(C) = 1$ exactly when the interval of t values determined by (29.15) and $t \geq 0$ is nonempty. Explicitly, every index with $c_{k-1} < 0$ supplies a lower bound

$$t \geq -\frac{c_k}{c_{k-1}}, \tag{29.16}$$

every index with $c_{k-1} > 0$ supplies an upper bound

$$t \leq -\frac{c_k}{c_{k-1}}, \tag{29.17}$$

and $c_{k-1} = 0$ requires $c_k \le 0$. The first coefficient supplies $t \le -c_1$.

Remark 29.7. The rigid margin Δ_K and the completion index ν measure different properties. The sign of Δ_K decides $q = 0$ feasibility, but its magnitude need not determine how many auxiliary degrees are necessary. The appropriate comparison is therefore not only $-\Delta_K$ versus ν, but the full scaled defect profile

$$E_0, E_1, \dots, E_\nu = 0.$$

For the eight $K = 7$ configurations in Experiment 28.1, this profile can be computed from the already estimated phase sets without any new frequency estimation.

Experiment 29.8. For each of the eight evaluable $K = 7$ phase sets, form the fixed core polynomial $C = P_{b,\Omega}$ once. Starting at $q = 0$, solve (29.8) in increasing degree. For every infeasible q, retain a dual vector satisfying (29.10); at the first feasible degree retain the multiplier coefficients, the completed Leslie coefficients, the minimum coefficient, the primal residual, and the spectral residual. Report

$$(b, \Delta_7, j_*, \nu, E_0, E_1, \dots, E_\nu)$$

for every case. This protocol certifies the minimal auxiliary repair degree rather than merely producing one high-dimensional feasible completion.

30. Decision criterion for the Leslie programme

Conjecture 30.1. *There exists a fixed K, a fixed or predeclared base rule, and a low-complexity transport law on $\mathcal{M}_K \cap \mathbb{R}^m_+$ such that consecutive rigid Leslie matrices inferred from past Mertens windows satisfy small transport residuals and predict future windows better than the null models without recalibration.*

Remark 30.2. The conjecture is falsified for a chosen protocol if rigid feasibility occurs only sporadically, mode matching is unstable, finite increments repeatedly leave the admissible set, velocity residuals are no smaller than in null sequences, or the transported model fails to improve future-window prediction.

Remark 30.3. A positive result would require all of the following: persistent rigid feasibility, coherent branch tracking, compatibility with the corrected cone (20.4), a transport law fixed from past data, and reproducible out-of-sample improvement. Static feasibility alone is insufficient.

31. Conclusions

Proposition 31.1. *The moving-window formulation has three exact layers. First, rigid Leslie spectra possess the strain geometry, positivity constraints, and modal transport developed above. Second, for fixed b their coefficient vectors lie in the K-dimensional reciprocal–Perron affine hull* (10.4)*. Consequently overlapping order-m recurrence windows satisfy the exact kernel law* (11.4)*; the first generic moving stride is $K + 2$; and at that stride the update is reduced to one scalar innovation, after which the remaining $K + 1$ new recurrence values are causally determined. Third, when a rigid core is infeasible, its minimal positive spectral repair is the first feasible Toeplitz linear program* (29.8)*, with exact dual infeasibility certificates* (29.10)*. These statements are algebraic and do not assume an arithmetic law for Mertens.*

Remark 31.2. The rigid $K = 7$ autoconsistency test is already negative on every evaluable grid point in Experiment 28.1, with a substantial negative margin. Therefore the rigid moving model cannot be invoked there before repair: auxiliary spectral states are necessary whenever a positive completion exists. The immediate numerical task is the certified minimal-completion protocol of Experiment 29.8. Only after identifying the minimal repaired cores does it make sense to test a moving-window transport law. For any repaired moving model that reaches the critical-stride regime, the remaining irreducible arithmetic quantity is again the scalar innovation τ_n; Leslie geometry alone does not determine it.

Notebook History

Version 1.0: static rigid Leslie family and spectral comparison.
Version 2.0: positive spectral completion with auxiliary eigenvalues.
Version 3.0: moving rigid Leslie matrices on consecutive windows, corrected tangent-cone formulation, first- and second-order spectral transport, and a falsifiable forecasting protocol.
Version 4.0: exact strain coordinates, minimum Perron value, universal critical region, phase Jacobian, viability conditions, and modal amplitude transport.
Version 5.0: fixed-Perron reciprocal affine hull, exact overlap rigidity, critical stride $K+2$, scalar innovation transport, admissible innovation intervals, direct determinant tests, and the closed $K = 1$ model.
Version 6.0: negative rigid $K = 7$ margin test, exact minimal auxiliary-completion index, Toeplitz LP formulation, Farkas certificates of lower-degree impossibility, and the monotone auxiliary defect profile.

References

[1] L. Benvenuti, *The inverse eigenvalue problem for Leslie matrices*, Electronic Journal of Linear Algebra **35** (2019), 319–330. DOI: 10.13001/1081-3810.3980.
[2] P. Duarte and S. Klein, *Lyapunov Exponents of Linear Cocycles: Continuity via Large Deviations*, Atlantis Studies in Dynamical Systems, vol. 3, Atlantis Press, 2016. DOI: 10.2991/978-94-6239-124-6.
[3] N. Yener, *On derivatives of eigenvalues, eigenvectors and generalized eigenvectors of matrices*, International Journal of Algebra **17** (2023), 61–75. DOI: 10.12988/ija.2023.91740.
[4] S. J. Wright, *Nonlinear Optimization 2: Lecture Notes*, University of Wisconsin–Madison, 2019.